

Classification Advanced

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Outline

-  Multiclass Classification
-  Rare class learning
-  Ensemble method

Multiclass Classification

- Binary classification → yes/no problem
- Multiclass classification → n different possibilities
- Diverse classification algorithms natively can be used for multiclass classification
 - Decision tree
 - Naïve Bayes Classifiers
- Logistic regression and SVM are natively binary classifiers
 - *Ad hoc* strategies can be defined to extend their behaviour in multiclass context

Multiclass Classification

- One-against-rest strategy
 - k classifiers are trained, k_i is a binary model that predicts, as positive class, if a given sample belongs to C_i class
 - At test time, if k_i correctly predicts C_i , the belonging to C_i gains a vote, otherwise a vote is given to other classes
 - The output is the label which obtained the majority of votes

Multiclass Classification

- One-against-one strategy
 - $\binom{k}{2}$ classifiers are trained, for each pair of classes
 - At test time, generic model $C_i\text{-against-}C_j$ provides a vote to the class predicted correctly
 - The output is the label which obtained the majority of votes
 - More classifiers with smaller training sets, thus resulting convenient from computational perspective

Rare class learning

- Classification of an unbalanced data sets, in which samples may belong to a rare class, is interesting since:
 - From domain perspective, rare class is *interesting* and *meaningful*
 - A classification error for the rare class is *critical*
 - *Weighting* and/or *sampling* techniques are required for classifiers to manage the unbalancing

Rare class learning

- Weighting

- Each sample has a weight connected to the cost of its mislabelling classification → favor rare classes
- The focus is towards maximizing the cost-weighted accuracy
 - Each sample belong to the i -th class, is associated to a cost $C(i)$ which is weighted while assigning the predicted label
- Easy to implement in already-existing classification algorithms
- The data set remains unchanged

Rare class learning

- Weighting

- Decision tree

- The weight is incorporated into the split criterion as Gini index or Entropy
 - It can also be added in pruning techniques

- Naive Bayes

- The estimation of the priors and probabilities of the features is weighted

- Support Vector Machines with slack variables

- Margin violation penalties are weighted for different classes

- k-NN

- Votes with respect to the individual classes of the neighboring k are weighted

Rare class learning

- Sampling

- Probability for a data point to be sampled is associated to its misclassification cost
- Oversampling of rare class
 - Can be harmful because it is obviously a replacement technique and therefore generates multiple copies of the same sample and may lead to overfitting
- Underampling over numerous classes
 - Usually preferred, training set is reduced
- Methodologies can be mixed

Rare class learning

- Sampling
 - Sampling with replacement
 - Sampled data point *can* be sampled again (duplicates are allowed)
 - Sampling without replacement
 - Sampled data point *cannot* be sampled again

Rare class learning

- Sampling
 - Synthetic Oversampling: SMOTE
 - The k neighbors of a rare sample, belonging to the minority class, are selected
 - A random fraction of selected samples is chosen, depending on the required level of oversampling
 - A synthetic sample is generated at a random point along the line joining the sample with each of the selected neighbors

Ensemble methods

- Ensemble methods combine predictions obtained from a set of algorithms, called base learners, over the data set
- Peculiar characteristics of each algorithm are involved to increase the overall accuracy

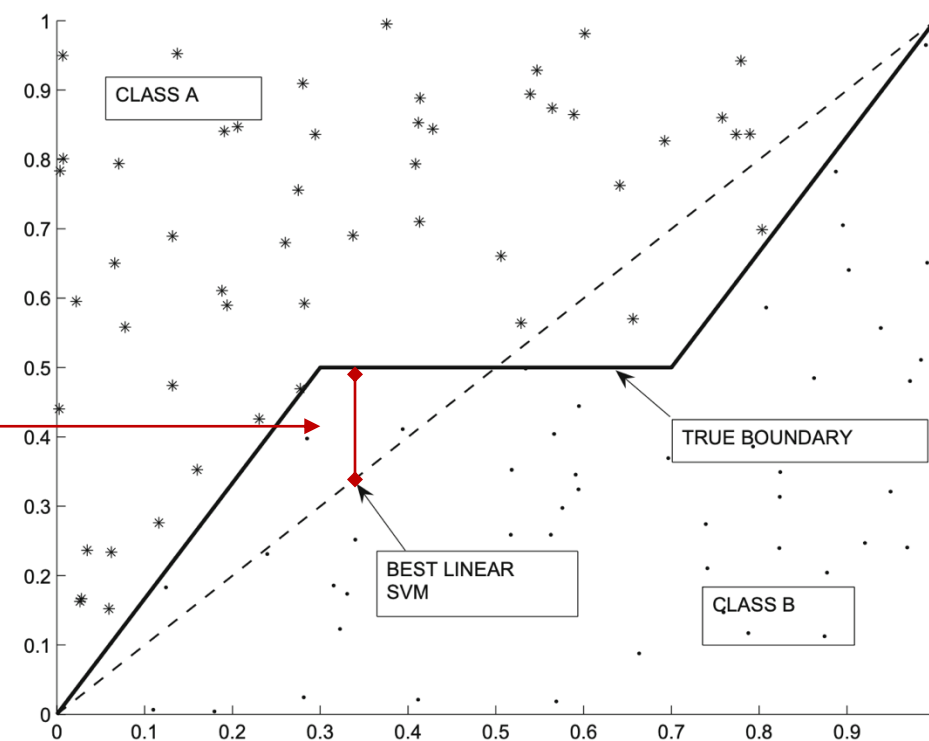
Algorithm *EnsembleClassify*(Training Data Set: $\mathcal{D}$
Base Algorithms: $\mathcal{A}_1 \dots \mathcal{A}_r$, Test Instances: $\mathcal{T}$)
begin
 $j = 1$;
 repeat
 Select an algorithm $\mathcal{Q}_j$ from $\mathcal{A}_1 \dots \mathcal{A}_r$;
 Create a new training data set $f_j(\mathcal{D})$ from $\mathcal{D}$;
 Apply $\mathcal{Q}_j$ to $f_j(\mathcal{D})$ to learn model $\mathcal{M}_j$;
 $j = j + 1$;
 until(termination);
 report labels of each $T \in \mathcal{T}$ based on combination of
 predictions from all learned models $\mathcal{M}_j$;
end

Ensemble methods

- Ensemble methods data-oriented
 - $\mathcal{A}_1, \dots, \mathcal{A}_r$ are instances of the same model
 - $f_j(\mathcal{D})$ differs for each model
 - Sampling
 - Manipulation of features or labels
 - Focus on portions of data where the classification is incorrect
- Ensemble methods model-oriented
 - $\mathcal{A}_1, \dots, \mathcal{A}_r$ are instances of different models
 - $f_j(\mathcal{D}) \equiv \mathcal{D}$
 - Different models perform better in different portions of the data set

Ensemble methods

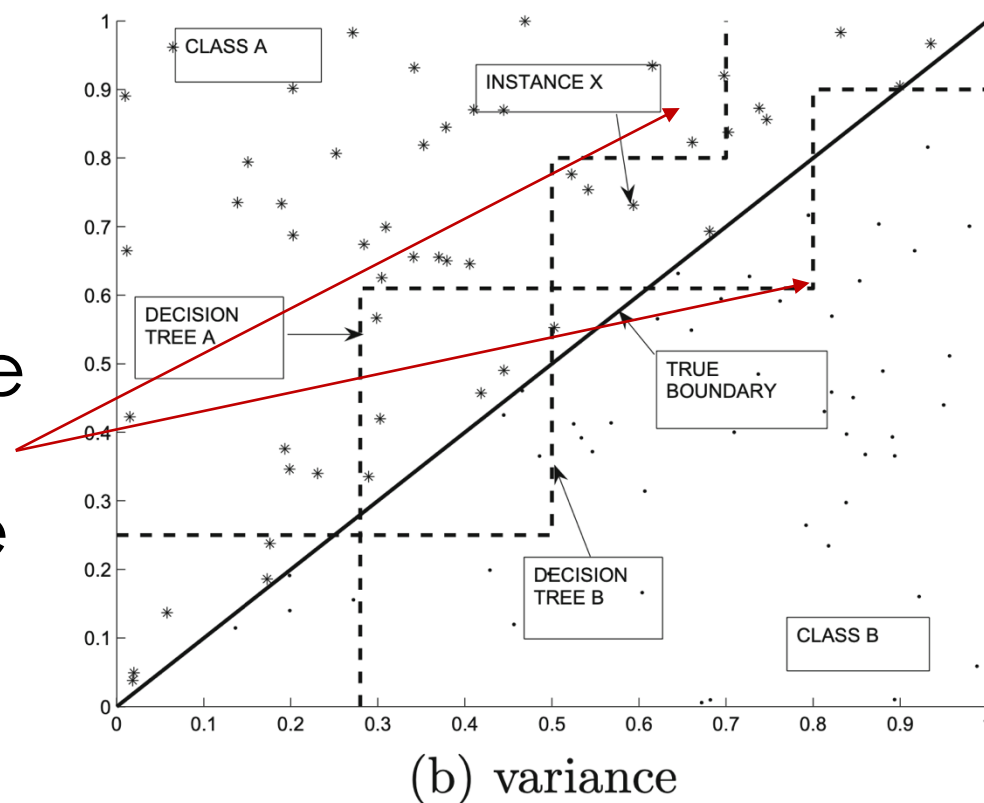
- Ensemble methods improve both the bias and variance of classifier estimation
- A single classifier exhibits a bias with respect to the real distribution of classes even for a large number of training data
 - SVM assumes data are lineary separable



(a) bias

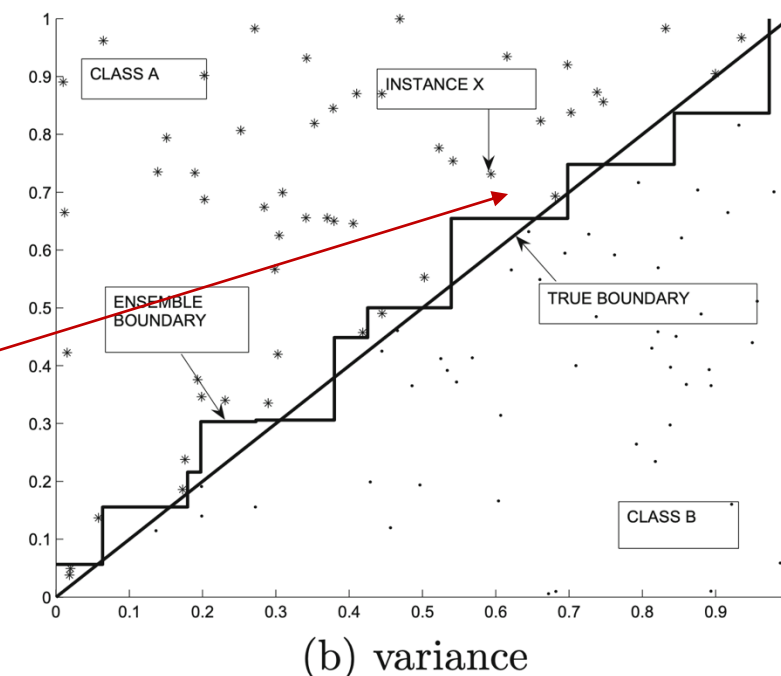
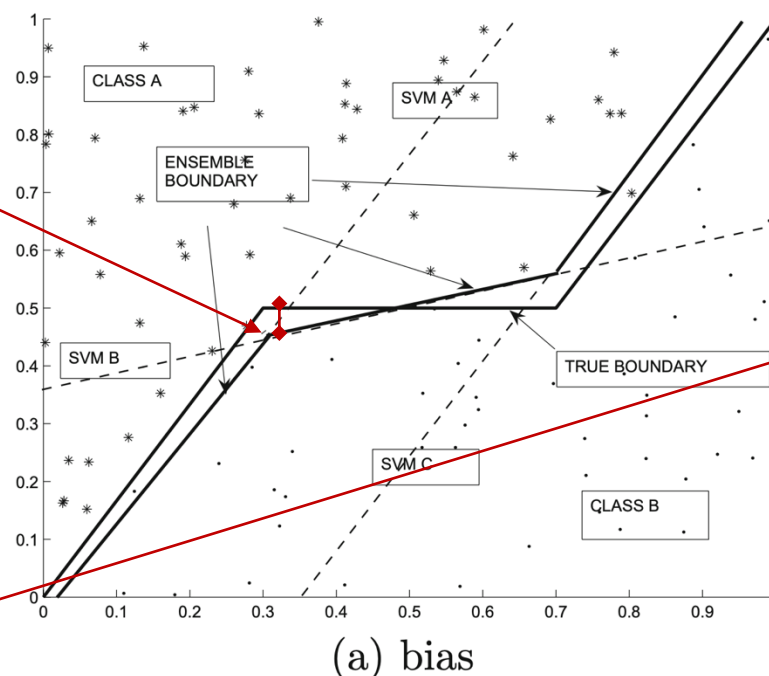
Ensemble methods

- Ensemble methods improve both the bias and variance of classifier estimation
- A single classifier has inherent variance in the prediction because as the instance of the training set changes, the prediction of the same sample can change



Ensemble methods

- Ensemble methods better approximates a complex boundary by reducing bias
- As the number of members of the ensemble increases, the estimate reduces its variance



Three independent decision trees

$$p_{tree,1} = p_{tree,2} = p_{tree,3} = p_{tree} = 80\%$$

$$p_{ensemble} = p_{tree,1} \times p_{tree,2} \times p_{tree,3} +$$

$$+ \binom{3}{2} p_{tree} \times p_{tree} \times (1 - p_{tree}) = \mathbf{89.6\%}$$

Ensemble methods

- Bagging (Bootstrapped aggregating)
 - Technique for creating an ensemble of i.i.d. classifiers in such a way as to reduce the variance of the estimation from σ^2 to σ^2/k
 - k data sets are generated via sampling with replacement
 - Obtained data sets have duplicates
 - It can be shown that the fraction of unique data points for each data set is $1 - (1 - 1/n)^n \approx 1 - 1/e$
 - This approach is called bootstrap
 - It is also used for calculating the accuracy of a model weighted on diverse instances slightly different of the data set

Ensemble methods

- Bagging (Bootstrapped aggregating)
 - k classifiers are trained and final prediction is given by the majority or mean of the expressed votes over each sample
 - Variance estimate is reduced, but not the bias
 - It is advisable to choose models that tend to reduce bias in themselves

Random Forests

- Random forests is an ensemble of decision trees trained with bagging
- The decision tree, especially if made to grow very deeply, greatly reduces bias even if at the great expense of the variance of the classification
- It is an ideal model for bagging

Random Forests

- The highest nodes in the hierarchy remain substantially invariant in the different trees
 - Individual trees are very correlated, which limits the reduction of estimation error
 - It is necessary to diversify the trees a lot during their growth to increase the prediction capacity of the ensemble
- Random Forests use a random criterion for splitting into the different trees of the ensemble

Random Forests

- Random split selection
 - Forest-RI (Random Input)
 - A random subset of q features is extracted for split
 - Forest-RC (Random Combination)
 - A subset L of features is extracted and q linear combinations are created that are used as multivariate features for the split
 - Preferable for low dimensionality of the data for which q becomes very small

Boosting

- Boosting is an iterative ensemble technique geared towards reducing bias
- The same learner is retrained several times on the data set where each sample is weighted with a higher weight if it was mis-classified at the previous iteration
- The classifier at iterate t is driven by the errors made by previous iterations

Boosting

- AdaBoost
 - Prediction is calculated as the weighted sum of predictions at each iterate t by the coefficient α_t
 - Variance can be reduced via resampling the data set with each iteration
 - Not suitable for noisy data

Boosting

- XGBoost (eXtreme Gradient Boosting)
 - Ensemble algorithm for classification and regression tasks
 - It is based on the concept of gradient boosting
 - Key idea:

Boosting can be interpreted as the minimization of a specific cost function in the functional space of the models

https://xgboost.readthedocs.io/en/release_3.0.0/install.html